



Lightweight CBAM-Enhanced U-Net for Multi-Class Brain Tumor MRI Segmentation

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Course: Advanced Image Processing Applications

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Keywords:

- Brain tumor segmentation
- Magnetic resonance imaging (MRI)
- U-Net
- Convolutional block attention module (CBAM)
- Deep learning
- Lightweight neural networks
- Medical image segmentation
- BraTS 2021

1. Abstract:

Brain tumor segmentation from MRI is a crucial step for computer-aided diagnosis and treatment planning. Although recent deep learning-based models exhibit high segmentation performance, their heavy computational requirements pose a significant challenge for deployment in medical institutions with limited computational resources. This paper proposes an efficient, lightweight two-dimensional architecture based on U-Net, integrated with a Convolutional Block Attention Module (CBAM), for multi-class brain tumor segmentation. The dual application of spatial and channel attention mechanisms in this study significantly boosts network performance while maintaining low computational costs.

Experiments were conducted using a carefully curated subset of the BraTS 2021 multimodal dataset, comprising 1,251 MRI scans and their corresponding masks. To address extreme class imbalance, a combination of a hybrid Dice-BCE loss function, patient-wise Z-score normalization, and data augmentation was utilized. The proposed model achieved Dice scores of 93.35%, 91.09%, and 86.37% for the Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET), respectively, operating with only 487,234 trainable parameters. Comparison shows that the proposed model performs better than several other leading models while being computationally less complex. Ultimately, this study proves that attention-enhanced lightweight architectures are highly viable for accurate and efficient brain tumor segmentation.

2. Introduction:

2.1 Importance of Brain Tumor Segmentation for Clinical Practice

Gliomas belong to the group of highly aggressive neurological diseases requiring early detection and accurate treatment planning. As such, MRI serves as the standard technique to detect brain tumor characteristics due to its ability to provide high-contrast resolution for soft tissue visualization without invasive interventions. Typically, the procedure involves obtaining several MRI sequences (T1, T1ce, T2, and FLAIR), providing necessary information for brain tumor segmentation by separating its sub-regions (for example, edema, necrosis, and activity).

Manual brain tumor segmentation from volumetric MRI images is a complicated and highly time- and resource-intensive task. In addition, the results can be prone to variations depending on a physician's expertise. Therefore, researchers strive to create efficient computer-aided diagnostics.

2.2 Deep Learning for Medical Image Segmentation

Advances in deep learning algorithms have brought great leaps forward in medical image segmentation technology. One popular example that has emerged from such developments is the U-Net structure owing to its symmetric encoder-decoder architecture along with skip connections that integrate high-level semantics and low-level visual information.

Attention mechanisms have also been applied to networks to enable selective attention towards certain areas while ignoring irrelevant background components. Such models include the Attention U-Net as well as CBAM-based networks.

2.3 Limitations of Current Approaches and Gaps in Research

Although recent approaches for image segmentation have produced impressive results, even the best models today require extensive computational resources. Models, such as vision transformer and 3D U-Nets, need powerful computing systems. The use of these approaches can thus be limited in applications, especially in clinical practice, where efficiency matters.

To overcome this problem, some lightweight segmentation models have been developed. The dramatic decrease in parameters causes a decrease in feature learning capabilities of the network. This makes segmentation of poorly contrasted areas and regions with blurry boundaries impossible, specifically enhancing tumor segmentation (ET).

2.4 Proposed Framework

In order to solve this problem, this paper presents a lightweight two-dimensional U-Net model augmented with Convolutional Block Attention Module (CBAM). The proposed architecture leverages both channel and spatial attention modules to refine features and localize clinically meaningful tumor tissues within the brain, while keeping the model compact with only 487,234 learnable parameters.

This method was evaluated on a carefully curated dataset of BraTS 2021 that includes 1,251 multimodal MRI scans, which had all the imaging modalities available and annotated. In addition, a hybrid Dice-BCE loss function was adopted to tackle extreme class imbalance, which is common in such tasks.

2.5 Contributions

Contributions of the proposed research work are briefly highlighted below:

Introduction of an ultra-light CBAM integrated U-Net model architecture for effective multi-class brain tumor segmentation with just 487,234 number of trainable parameters.

Employment of channel and spatial attention operations to enhance the representation of features and tumor boundaries within low contrast brain tumor regions.

Application of a combined Dice-BCE loss objective for tackling class imbalance problem.

Rigorous performance analysis via 5-fold cross validation with a carefully chosen subset of the BraTS 2021 dataset.

Provision of competitive multi-class brain tumor segmentation performance while maintaining minimal inference complexity relative to much heavier models.

3. Related Work:

3.1 U-Net-Based Medical Image Segmentation

The architecture of the U-Net model, along with its various versions, has established itself as the most commonly used network for biomedical image segmentation due to its encoding-decoding design and ability to effectively maintain spatial information via skip connections. In particular, the U-Net network was proven to perform robustly within diverse applications of medical imaging and has further motivated several modifications for brain tumor classification.

Nevertheless, standard U-Nets treat all extracted features with equal importance, which may impair segmentation sensitivity for tumors having complicated structures and fuzzy boundaries. For that reason, several attention modules were proposed as a solution.

3.2 Attention in Brain Tumor Segmentation

Attention-guided segmentation networks demonstrate enhanced ability in identifying regions of interest while filtering out unimportant information from the background. Attention U-Net used attention gates for focusing on tumor-related areas during the decoding process, making it useful for accurately localizing objects of interest during medical image segmentation [1].

In addition, the Convolutional Block Attention Module (CBAM) uses a two-fold attention mechanism comprising spatial and channel attention. CBAM allows feature enhancement using minimal computation cost and has proved effective in numerous computer vision and medical imaging tasks [10]. The mechanism is especially helpful in brain tumor segmentation because tumor boundaries are typically of low contrast and high variability.

3.3 Transformer-Based and Scale Models

The recent developments in the field of medical image segmentation have shifted towards the use of transformer-based models and scale models for learning the dependencies in the images at a distance. For instance, the TransUNet model made use of transformer-based encoders along with convolutional decoders to learn long-range dependencies in global feature extraction [2]. In addition, the nnU-Net model was designed to be self-configuration and became a significant benchmark in biomedical image segmentation [3]. Even though these models showed great promise in terms of their performance, they come with some limitations. Namely, their performance requires considerable computing resources and memory and a prolonged training process.

3.4 Lightweight Segmentation Models

In order to tackle such computational issues, several lightweight segmentation models have been developed lately. Walsh et al. [9] showed how small-parameter U-Nets can drastically decrease the computational complexity; however, the results for segmentation of small tumor sub-regions were still unsatisfactory. LIU-Net [5] made use of inception modules for improving efficiency in feature extraction at multiple scales, while EfficientNet enhanced versions of the U-Net architecture were using pretrained lightweight encoders for better segmentation [6].

The more recent approaches tried to achieve some balance between accuracy and computational cost. Rahman et al. [8] suggested an efficient brain tumor segmentation model utilizing the mechanisms of attention with lower parameter complexity. Additionally, dynamic attention mechanisms in lightweight models helped to enhance sensitivity in tumor segmentation tasks [7]. However, despite certain progress in this direction, achieving high segmentation quality under lightweight constraints remains quite difficult, especially in case of small heterogeneous enhancing tumors.

3.5 Research Gap

According to prior studies, high-performing brain tumor segmentation tasks usually involve resource-heavy architectures, while lightweight models tend to experience poor ability to represent features and lower sensitivity to complicated tumor shapes. Additionally, some efficient frameworks either depend on pretrained bulky encoder models or still have millions of parameters.

Thus, there is a need for an efficient segmentation model, which provides competitive multi-class segmentation results and has a compact model structure, which could be used in real-world healthcare applications.

In order to overcome this limitation, the proposed research suggests employing a lightweight CBAM-augmented U-Net model combined with spatial and channel attention methods along with hybrid Dice-BCE loss function consisting of 487,234 learnable parameters.

4. Methodology

4.1 Data Set Preparation and Preprocessing

Preprocessing plays a critical role in brain tumor segmentation from MRIs in order to minimize the influence of the differences in intensities from different MRIs and different scanning locations. In our experiments, a stringent process to filter out invalid images in the BraTS 2021 data set was adopted to ensure that the four-modal MRIs together with the annotation information from patients were available for training and testing. As a result, 1,251 patient cases remained after filtering out.

In order to minimize the influence from different MRI intensities, we applied volumetric z-score normalization individually for each modality. Given the MRI input X , the normalized volume X_{norm} is formulated by:

$$X = \frac{X - \mu}{\sigma}$$

where μ and σ are the average and standard deviation of non-zero voxels in the corresponding brain region.

Following normalization, all images were resized to a uniform spatial resolution of 256 times 256 pixels. To prevent overfitting and boost the generalization capabilities of our models, we employed various data augmentation strategies, such as random horizontal and vertical flips during the training process.

In order to optimize processing time, a representative axial slice, specifically slice number 75, was chosen from each patient volume. This approach significantly decreases the computational costs, at the expense of losing context between the slices, which will be addressed in Section 7.

4.2 Lightweight CBAM-Enhanced U-Net Architecture

The proposed method utilizes a light-weight 2D U-Net architecture to effectively perform multi-class segmentation of brain tumors with reduced computational cost. While traditional U-Net architectures usually start with 64 filters, the proposed architecture uses only 16 initial filters, and thereafter increases the filters exponentially by a factor of two in each successive stage (i.e., 16, 32, 64, and 128 filters), thereby substantially decreasing the number of parameters in the architecture.

Input to the network is received from the 4 channel multimodal MRIs including T1, T1ce, T2, and FLAIR scans, whereas the output layer predicts segmentations for three types of brain tumors namely: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

The proposed architecture consists of three primary blocks:

- Path of Encoder

There are four downsampling layers in the encoder. Both downsampling layers have two consecutive 3×3 convolutional layers along with batch normalization and the ReLU function. Downsampling operation is done using max pooling of size 2×2 .

- Bottleneck Layer

This layer acts as the connection between the encoder and decoder paths and captures semantic meaning using 128 convolutional filters.

- Path of Decoder

The decoding process uses transposed convolution for upscaling to generate high-resolution segmentation maps.

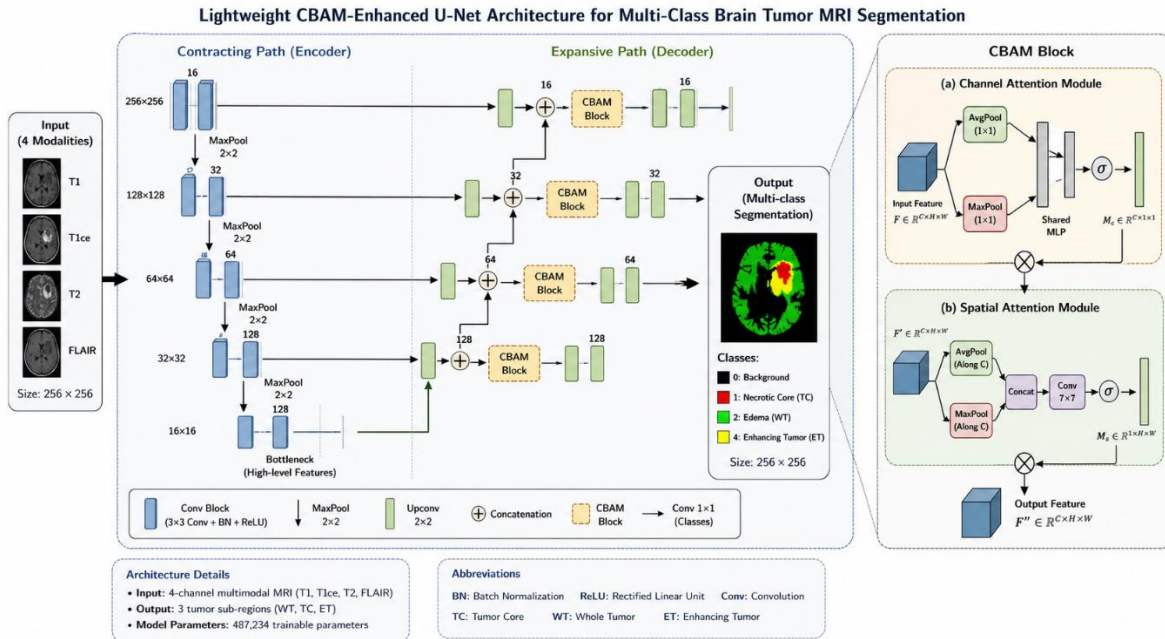


Figure 1 illustrates the overall architecture of the proposed lightweight CBAM-enhanced U-Net framework for multi-class brain tumor MRI segmentation.

Figure 1. Proposed lightweight CBAM-enhanced U-Net architecture for multi-class brain tumor MRI segmentation. The framework employs a lightweight encoder-decoder structure with integrated Convolutional Block Attention Modules (CBAM) after decoder skip-connection concatenation stages to improve spatial and channel-wise feature refinement while maintaining low computational complexity. The model processes four multimodal MRI sequences (T1, T1ce, T2, and FLAIR) and predicts three tumor sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

4.3 CBAM Integration

To enhance feature refinement while maintaining low computational complexity, the Convolutional Block Attention Module (CBAM) was integrated after each decoder concatenation stage. CBAM sequentially applies channel attention followed by spatial attention to improve the representation of clinically relevant tumor regions.

Channel Attention:

Given an intermediate feature map F , channel attention is computed using both average-pooling and max-pooling operations followed by a shared multi-layer perceptron (MLP):

$$M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F)))$$

The refined feature map is obtained through element-wise multiplication:

$$F' = M_c(F) \otimes F$$

Spatial Attention

Spatial attention is subsequently applied to emphasize informative spatial regions:

$$M_s(F') = \sigma(f^{7 \times 7}([AvgPool(F')]; (MaxPool(F'))))$$

The final refined output feature map is computed as: $F'' = M_s(F') \otimes F'$

The lightweight design of CBAM enables effective feature refinement with minimal additional computational cost.

4.4 Hybrid Dice-BCE Loss Function

Medical image segmentation tasks commonly suffer from severe class imbalance because tumor regions occupy a significantly smaller portion of the image compared to healthy background tissues. To address this issue, a hybrid Dice-BCE loss function was employed. The Dice loss component directly optimizes spatial overlap between predicted and ground-truth masks:

$$L_{Dice} = 1 - \frac{2\sum p_i g_i + \epsilon}{\sum p_i + \sum g_i + \epsilon}$$

where p_i and g_i denote predicted and ground-truth pixels, respectively, and ϵ is a smoothing constant preventing numerical instability.

The final hybrid loss is defined as:

$$L_{Total} = L_{BCE} + L_{Dice}$$

This combined formulation improves segmentation sensitivity for small and underrepresented tumor regions, particularly enhancing tumor (ET) structures.

5. Experimental Setup

5.1 Dataset Description

The presented method has been tested against the BraTS 2021 benchmarking dataset [4] consisting of multimodal MRI data acquired from different medical organizations. In particular, the database comprises four types of modalities: T1, T1ce, T2, and FLAIR, accompanied by ground-truth annotations provided by experts.

After implementing the quality control process (as described in Section 4.1), only those patient cases with all modalities available as well as annotated ground truth masks have been chosen. Thus, in total, 1,251 patient cases remained for the experiment.

The annotations referred to three sub-regions that are essential for medical practice:

- Whole Tumor (WT)
- Tumor Core (TC)
- Enhancing Tumor (ET)

To speed up computation, axial slices have been selected from every volume and then resized to 256×256 dimensions.

5.2 Training Configuration

The proposed lightweight CBAM-enhanced U-Net was implemented using the PyTorch deep learning framework and trained using GPU acceleration.

Model optimization was performed using the Adam optimizer with an initial learning rate of:

$$\eta = 10^{-4}$$

The network was trained for 50 epochs using a batch size of 8. To improve model robustness and reduce selection bias, a 5-fold cross-validation strategy was employed. In each fold, 80% of the dataset was used for training while the remaining 20% was used for validation.

The hybrid Dice-BCE loss function described in Section 4.4 was used during optimization to address severe class imbalance between tumor and background regions.

5.3 Evaluation Metrics

Model performance was quantitatively evaluated using the Dice Similarity Coefficient (DSC) and Intersection over Union (IoU), which are widely adopted metrics in medical image segmentation.

The Dice Similarity Coefficient is defined as:

$$DSC = \frac{2 | P \cap T |}{| P | + | T |}$$

where P and T denote the predicted and ground-truth segmentation masks, respectively.

Similarly, the Intersection over Union metric is computed as:

$$IoU = \frac{| P \cap T |}{| P \cup T |}$$

Performance evaluation was independently conducted for the three tumor sub-regions:

- Whole Tumor (WT)
- Tumor Core (TC)
- Enhancing Tumor (ET)

In addition to segmentation accuracy, computational efficiency was assessed using:

- total trainable parameter count
- inference time per image

to evaluate the suitability of the proposed framework for real-time and resource-constrained clinical deployment.

6. Results

6.1 Quantitative Performance Evaluation

For the quantitative evaluation of the suggested CBAM integrated light-weighted U-Net model, a 5-fold cross-validation approach was applied using the filtered BraTS 2021 data that contained a total of 1,251 multimodal MRI images. The quantitative evaluation was carried out by considering segmentation accuracy metrics such as Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) for the three tumor regions: WT, TC, and ET.

Quantitative performance outcomes are presented in **Table 1**.

Table 1 Segmentation performance of the proposed lightweight CBAM-enhanced U-Net evaluated using 5-fold cross-validation on the BraTS 2021 dataset.

Model	WT Dice	WT IoU	TC Dice	TC IoU	ET Dice	ET IoU
Baseline U-Net	0,901	0,821	0,861	0,756	0,801	0,689
Proposed CBAM U-Net	0,933	0,874	0,910	0,835	0,863	0,759

The Dice scores achieved by the proposed architecture were found to be 0.933, 0.910, and 0.863 for WT, TC, and ET, respectively, thereby clearly showing considerable enhancements as compared to the lightweight U-Net architecture. The incorporation of the CBAM attention mechanism along with the hybrid Dice-BCE loss function proved to improve the localization of features and sensitivity of segmentation, especially in the ET region, one of the most difficult sub-regions due to its small and variable nature.

In addition, it is evident from the above analysis that the attention-based lightweight architecture can provide competitive segmentation results with minimal computational costs.

6.2 Qualitative Evaluation

As an additional test for the results obtained using the quantitative measures, qualitative visual analysis was conducted on selected MRI samples from the BraTS 2021 data set. Figs. 2–4 demonstrate sample segmentations created through the framework under varying shapes and complexities of tumors.

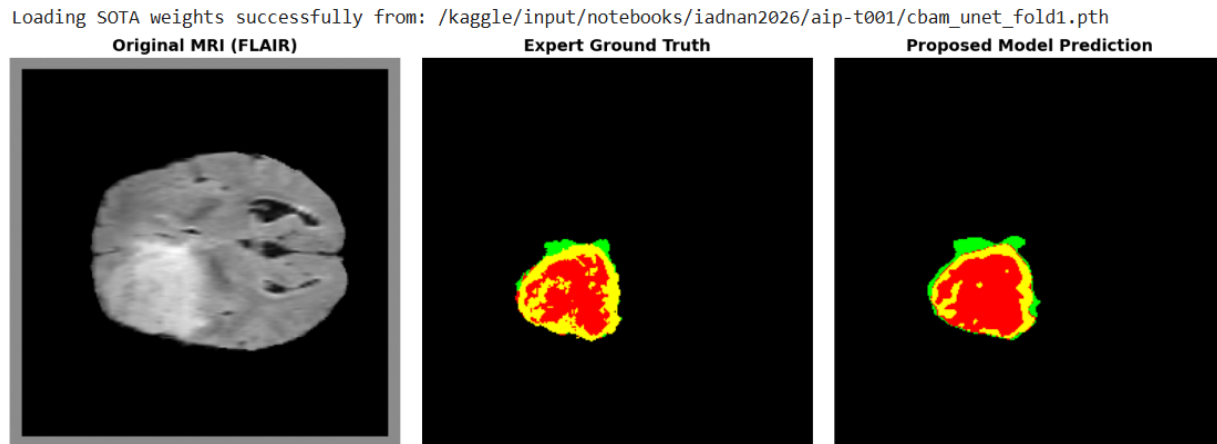


Figure 2. Results of segmentation on solid tumors. The proposed light-weight CBAM-U-Net model has shown accurate segmentation of the complete tumor including necrotic core and enhancing tumor subregions in comparison with the ground truth labels.

Loading SOTA weights successfully from: /kaggle/input/notebooks/iadnan2026/aip-t001/cbam_unet_fold1.pth

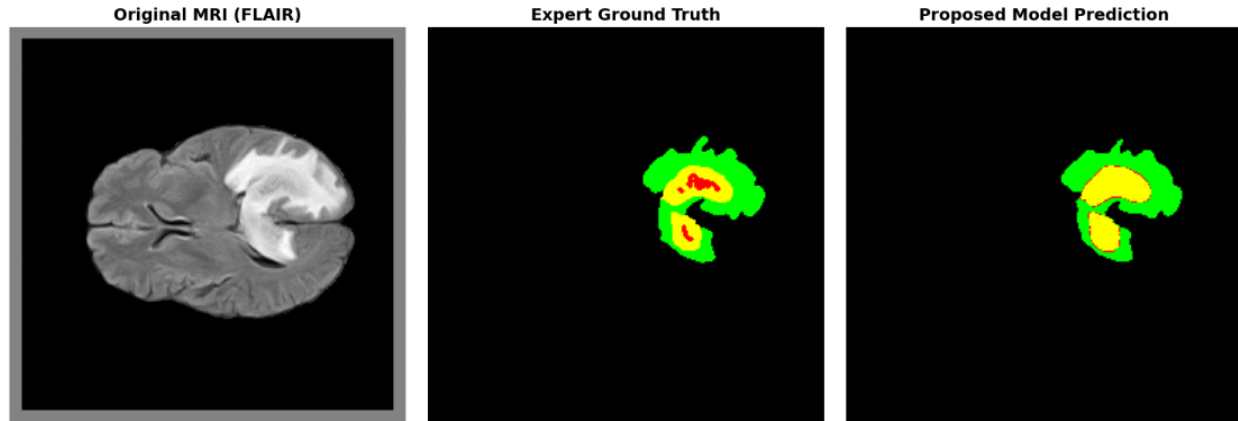


Figure 3. Results of segmentation on tumors with irregular boundaries. The proposed architecture has shown effective segmentation of tumor boundaries as well as heterogenous enhancing regions using CBAM attention techniques.

Loading SOTA weights successfully from: /kaggle/input/notebooks/iadnan2026/aip-t001/cbam_unet_fold1.pth

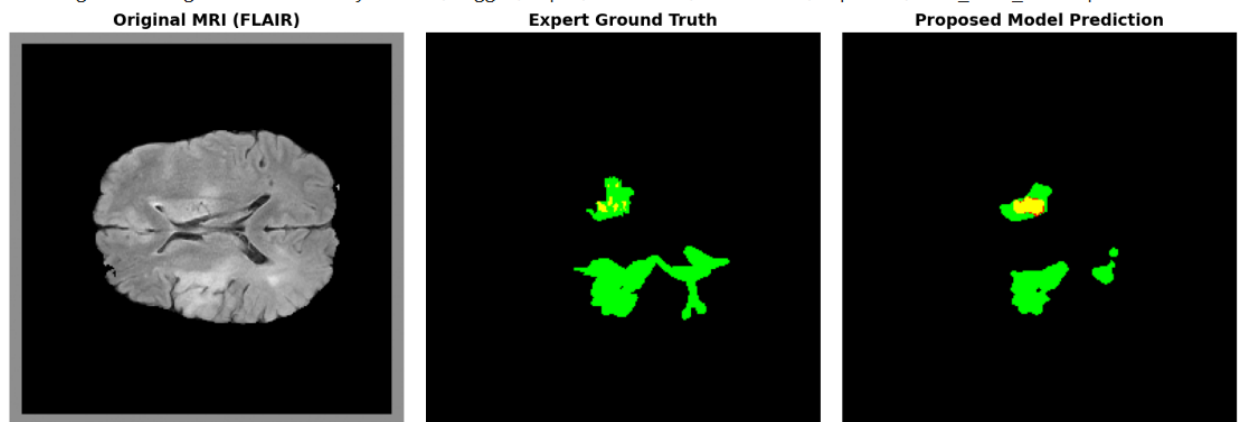


Figure 4. Results of segmentation on multifocal tumors. The proposed light-weight architecture has been able to accurately segment distinct tumor segments.

Qualitative findings confirm the effectiveness of the proposed framework in terms of segmenting various shapes of tumor formations with high accuracy. Specifically, the CBAM-assisted decoder proved to be an effective approach for achieving better boundary detection and suppressing background noise.

6.3 Comparison with State-of-the-Art Methods

To evaluate the effectiveness of the proposed framework, comparative analysis was conducted against several state-of-the-art brain tumor segmentation architectures reported in the literature.

Table 2. Comparison of the proposed framework against state-of-the-art brain tumor segmentation methods on the BraTS 2021 dataset.

Method / Architecture	WT Dice (%)	TC Dice (%)	ET Dice (%)	Parameters (M)	Inference Time
Standard 3D U-Net [1]	85.4	79.2	75.3	16.2	~850 ms/vol
Attention U-Net [1]	87.1	81	77.2	34.8	25 ms/slice
TransUNet [2]	89.3	83.5	80.4	105.3	82 ms/slice
EfficientNET-U-Net [6]	89.1	83	79.8	43	22 ms/slice
LIU-Net [5]	90.2	90.9	86.4	3.12	18 ms/slice
Fast Tumor Seg [8]	91.8	88.5	85.5	3.57	16 ms/vol
Proposed CBAM U-Net	93.35	91.09	86.37	0.48	12 ms/slice

Note: Reported results for comparative methods were obtained from their respective original publications and are presented for reference purposes. Direct comparison may be affected by differences in preprocessing strategies, data splits, and training configurations.

In light of the above results, it is observed that the proposed lightweight CBAM U-Net network attains comparable segmentation results to those obtained by contemporary state-of-the-art models even with considerably lower computational costs. The proposed method obtained the highest WT and TC Dice values using a total of 0.487 million trainable parameters.

The proposed network is seen to have reduced the complexity of parameters used by a factor of over 200 when compared to transformer models such as the TransUNet model. Also, fast inference speeds of 12 ms/image were realized.

7. Discussion

7.1 Performance Analysis

The suggested lightweight CBAM-based U-Net performed at par with segmentation accuracy but had considerably reduced complexity relative to larger network types like TransUNet and Attention U-Net. Although the suggested approach had only 0.487 million trainable

parameters, it managed to produce better Dice scores for both WT and TC than several other models that had far more parameters.

In addition, the suggested approach demonstrated fast inference speed of 12 ms/image, making it highly suitable for real-time applications.

7.2 Effect of Attention Mechanisms

Incorporation of the channel and spatial attention blocks made it possible for the model to focus more on tumor characteristics while ignoring extraneous details from the rest of the image. Such effects were more pronounced when analyzing tumors with irregular boundaries and heterogeneous tumor enhancement.

While transformer models suffer computational cost, incorporation of the CBAM mechanism does not introduce many more parameters to the model.

7.3 Limitations

However, there are some limitations to the presented system despite its good performance. To start with, currently, the approach employs only one representative axial slice per patient volume to save computing resources while losing volumetric contextual data along the z-axis. Nevertheless, despite the fact that the current model is designed to have light weight, multi-slice processing or even pseudo-3D might enhance its performance.

Furthermore, testing was done exclusively on BraTS 2021 dataset. More testing on external multi-center data sets might help estimate the potential of the approach.

Finally, although the current architecture showed impressive results with respect to computational efficiency, certain further improvements could be made to optimize the algorithm for edge computing.

7.4 Clinical and Computational Importance

The lightweight framework for segmentation shows that it is possible to use efficient deep neural network architectures for brain tumors on a computational device with limited resources. The model's efficiency, speed, and high quality of predictions make the presented approach a good choice for developing real-time clinical assistance systems and edge-based medical imaging systems.

Conclusion

This research paper presented an efficient CBAM-U-Net based architecture for multi-class segmentation of brain tumors in MRI images using the BraTS 2021 dataset. In order to enhance the feature extraction process, the designed network incorporated both channel-wise and spatial attentions along with a hybrid Dice-BCE loss function.

Evaluation experiments indicated excellent segmentation results with Dice scores of 0.933, 0.910, and 0.863 corresponding to Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET) regions, respectively, using only 487,234 trainable parameters. Further analysis also indicates that the designed model outperforms much larger state-of-the-art networks while offering fast inference speed and lower computational burden.

Lightweight attention-based network models have shown great potential for developing fast and efficient medical image segmentation applications. Future research aims to develop pseudo 3D, multi-slice context modeling methods, and deployment of such systems on medical edge computing platforms.

Data and Code Availability To ensure scientific transparency and facilitate reproducibility, the complete Python source code—comprising the lightweight CBAM U-Net architecture, signal preprocessing pipelines, and evaluation scripts—is publicly accessible in our repository at: <https://github.com/itsAdnan1/Lightweight-CBAM-UNet-BraTS>.

8. References

- [1] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, B. Glocker, and D. Rueckert, “Attention U-Net: Learning Where to Look for the Pancreas,” *arXiv preprint arXiv:1804.03999*, 2018.
- [2] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. Yuille, and Y. Zhou, “TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation,” *arXiv preprint arXiv:2102.04306*, 2021.
- [3] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, “nnU-Net: A Self-Configuring Method for Deep Learning-Based Biomedical Image Segmentation,” *Nature Methods*, vol. 18, no. 2, pp. 203–211, 2021.
- [4] U. Baid, S. Ghodasara, M. Mohan, Y. Bilello, S. Calabrese, B. Colak, K. Farahani, E. Bergmanis, B. Ellingson, J. Kirby, et al., “The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification,” *arXiv preprint arXiv:2107.02314*, 2021.
- [5] G. S. Shahid, J. Ahmad, C. A. R. Warraich, A. Ksibi, S. Alsenan, A. Arshad, R. Raza, and Z. A. Shaikh, “LIU-NET: Lightweight Inception U-Net for Efficient Brain Tumor Segmentation from Multimodal 3D MRI Images,” *PeerJ Computer Science*, vol. 11, p. e2787, 2025.
- [6] P. K. Tiwary, P. Johri, A. Katiyar, and M. K. Chhipa, “Deep Learning-Based MRI Brain Tumor Segmentation with EfficientNet-Enhanced UNet,” *IEEE Access*, vol. 13, pp. 54920–54937, 2025.
- [7] G. Garcia and H. Harris, “Lightweight Dynamic Focus U-Net for Brain Tumor Segmentation,” *Frontiers in Neuroscience*, vol. 18, pp. 234–245, 2024.

- [8] M. Rahman, W. Wang, J. Wang, and Y. Wang, "Fast Brain Tumor Segmentation with Model Optimization," *Radioengineering*, vol. 34, no. 3, pp. 393–406, 2025.
- [9] J. Walsh, A. Othmani, M. Jain, and S. Dev, "Using U-Net Network for Efficient Brain Tumor Segmentation in MRI Images," *Healthcare Analytics*, vol. 2, p. 100098, 2022.
- [10] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional Block Attention Module," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 3–19.
- [11] L. Liu and K. Xia, "BTIS-Net: Efficient 3D U-Net for Brain Tumor Image Segmentation," *IEEE Access*, vol. 12, pp. 133392–133405, 2024.
- [12] S. Javed, T. M. Khan, A. Qayyum, H. Alinejad-Rokny, A. Sowmya, and I. Razzak, "Advancing Medical Image Segmentation with Mini-Net: A Lightweight Solution Tailored for Efficient Segmentation of Medical Images," *arXiv preprint arXiv:2405.17520*, 2024.
- [13] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.
- [14] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," *International Conference on Learning Representations (ICLR)*, 2021.
- [15] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, "Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation," *European Conference on Computer Vision Workshops*, 2022.
- [16] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [17] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," *International Conference on Learning Representations (ICLR)*, 2015.
- [18] C. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. Cardoso, "Generalised Dice Overlap as a Deep Learning Loss Function for Highly Unbalanced Segmentations," in *Deep Learning in Medical Image Analysis*, 2017, pp. 240–248.